

Impact Statement

Our project addresses a foundational barrier to equitable participation in the AI ecosystem: the poor tokenization of many African languages. Existing tokenizers are predominantly optimized for high-resource languages such as English, causing African words—particularly those from morphologically rich and agglutinative languages—to be fragmented into excessive and often meaningless tokens. Our localized tokenization pipeline and SuperBPE approach aim to preserve meaningful linguistic units while reducing token fragmentation and computational overhead.

The potential positive impact extends across individuals, communities, researchers, businesses, and public institutions. More efficient tokenization could reduce the computational and financial cost of developing and deploying AI applications in African languages, making technologies such as education platforms, translation systems, healthcare assistants, and digital services more accessible. For native speakers, better preservation of words, morphology, diacritics, and other linguistic features could contribute to more accurate and culturally appropriate AI systems. In the longer term, improving African-language representation in AI could support greater digital inclusion and reduce linguistic disparities in access to emerging technologies.

However, the project also carries risks. A tokenizer trained on incomplete or unbalanced data could reinforce existing linguistic inequalities by performing better for some languages or dialects than others. Incorrect handling of morphology or orthographic features could also distort linguistic meaning and undermine cultural preservation. In addition, greater availability of African-language AI infrastructure could enable harmful applications if downstream systems are deployed without appropriate safeguards. Claims about efficiency or performance could also be misleading if they are based on limited language coverage or benchmarks.

We will mitigate these risks through balanced and transparent corpus selection, evaluation across different African language families, native-speaker and community feedback, quality audits, and open documentation of dataset limitations. Dataset metadata, preprocessing methods, and evaluation results will be made transparent so that researchers and communities can identify gaps and contribute improvements. Iterative updates will address vocabulary gaps and encoding errors as new evidence becomes available.

Our impact assessment remains subject to uncertainty because African languages are highly diverse, and improvements in tokenization do not automatically guarantee better downstream model performance. We therefore view SuperBPE as foundational infrastructure rather than a complete solution to African-language AI challenges. Long-term impact will depend on continued research, community participation, responsible deployment, and collaboration among researchers, developers, institutions, and native-language communities.